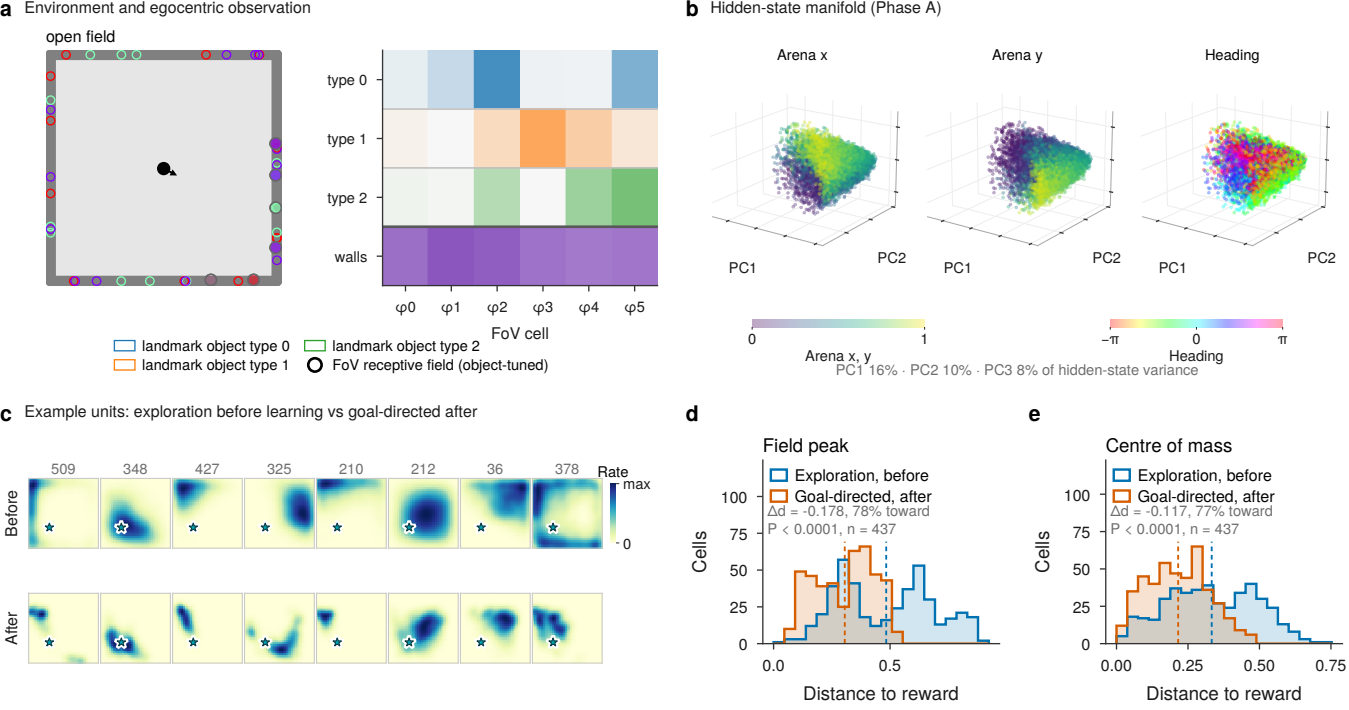

a
Environment and egocentric observation
open field
type 0
type 1
type 2
walls
φ0
φ1
φ2
φ3
φ4
φ5
FoV cell
landmark object type 0
landmark object type 1
landmark object type 2
FoV receptive field (object-tuned)
b
Hidden-state manifold (Phase A)
Arena x
Arena y
Heading
PC1
PC2
0
1
Arena x, y
−π
0
π
Heading
PC1 16% · PC2 10% · PC3 8% of hidden-state variance
c
Example units: exploration before learning vs goal-directed after
509
348
427
325
210
212
36
378
Rate
max
0
Before
After
d
Field peak
Exploration, before
Goal-directed, after
Δd = -0.178, 78% toward
P < 0.0001, n = 437
Cells
Distance to reward
e
Centre of mass
Exploration, before
Goal-directed, after
Δd = -0.117, 77% toward
P < 0.0001, n = 437
Cells
Distance to reward